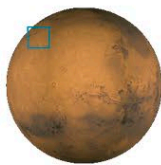


## THARSIS REGION

## Acheron Fossae



**TYPE** Fault system  
**AGE** Over 3.5 billion years  
**LENGTH** 695 miles (1,120 km)

Acheron Fossae is a relatively high area that has seen intense tectonic activity in the past. It marks the northern edge of the Tharsis Bulge



## CLIFF FACE

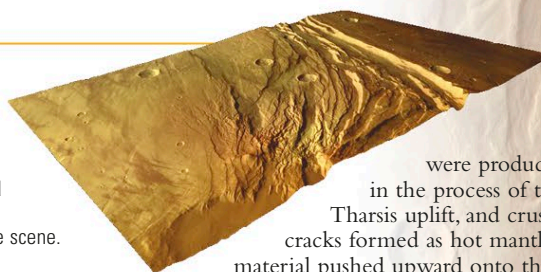
This is a close-up of the bright, steep slopes of a scarp or cliff. Dark streaks on the cliff face may be formed as the dust mantle that covers the region gives way and produces a dust avalanche.



## BROKEN CRUST

In this perspective view across the highly deformed area of Acheron Fossae, curved faults trending to the northwest dominate the scene.

and is located about 600 miles (1,000 km) north of Olympus Mons. Acheron is part of a network of fractures that radiates out from the Tharsis Bulge—a huge region of uplift and volcanic activity. It can be compared to the East African Rift on Earth (see p.130), where continental plates have spread apart. The huge curved faults in the Tharsis Bulge



were produced in the process of the Tharsis uplift, and crustal cracks formed as hot mantle material pushed upward onto the surface rock layer. The brittle Martian crust broke along zones of weakness when the tension of the uplift became too strong. The eroded walls, the subdued nature of the high-standing hills, and the wind etching on the flat surfaces all confirm this is ancient terrain. This type of feature gets its name from the Latin *fossa*, meaning “trough.” In Greek mythology, Acheron is the river that flows into Hades, the Underworld.

## GRABENS AND HORSTS

The planetary crust has fallen between parallel faults to form grabens up to 1 mile (1.7 km) deep; remnants of the preexisting heights are termed horsts.

## STRESSED LANDSCAPE

A fault system cutting across an ancient impact crater is evidence of the stress felt by the Martian crust. The crater floor has since been resurfaced by material from outside the area.

## CIMMERIA TERRA

## Apollinaris Patera



**TYPE** Patera volcano  
**AGE** 900 million years  
**DIAMETER** 184 miles (296 km)

This is an example of a type of volcano that was first identified on Mars. Known as patera volcanoes, they have very gentle slopes (with angles as low as 0.25°). Apollinaris Patera is one of the largest on the planet, situated on the northern edge of Cimmeria Terra, a few degrees south of the equator. It is the only



## MESAS AND TROUGHS

A group of mesas was created by pitting and erosion of the surface in an area north of Apollinaris Patera. Windblown dust has filled the troughs between the mesas.

major volcano that is isolated from the two major volcanic regions of Tharsis, to the northeast, and Elysium, to the northwest. Apollinaris is a

broad, roughly shield-shaped volcano, reminiscent of an upturned saucer. It is only about 3 miles (5 km) high and has a caldera about 50 miles (80 km) across. It appears to have been formed by both effusive and explosive activity. Lava flows are clearly visible beyond the summit. A cliff surrounding the caldera area is visible on the northern side but has disappeared on the opposite side. It is buried under a fan of material whose surface is marked by broad channels. The fan material could have formed from flowing lava or volcanic rock fragments.

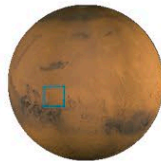
## SPLIT-LEVEL CALDERA

The caldera has two different floor levels. It is partially hidden here by a patch of blue-white clouds. The summit area is pocked with impact craters.



## AONIA TERRA

## Claritas Fossae



**TYPE** Fault system  
**AGE** Over 3.5 billion years  
**LENGTH** 1,275 miles (2,050 km)

Claritas Fossae is a series of roughly northwest-to-southeast-trending linear fractures, which forms the southern end of the Tharsis Bulge. It is located south of the equator at the western end of the Valles Marineris. The region is about 95 miles (150 km) wide at its northern end and 340 miles (550 km) wide in the south. Individual fractures range from a few to tens of miles across.

## FAULTS IN CLARITAS FOSSAE

Running from the volcanic Tharsis Bulge, the linear features in this image of Claritas Fossae coincide with fractures in the Martian crust produced by stretching forces.



## LAVA BLANKET

The eastern part of Claritas Fossae (bottom) meets the western part of Solis Planum (top). The lava from Solis has flowed over some of the older fractured terrain of Claritas and surrounds some of the higher ground.

They formed as a result of enormous stresses associated with the formation of the Tharsis Bulge. As the crust pulled apart, blocks of crust dropped between two faults to form features called grabens. Crustal blocks that remained in place or were thrown up are termed horsts. Claritas Fossae separates two volcanic plains: that of Solis Planum to the east and Daedalia Planum to the west.

